

SiyaBusa ERP

AI Capabilities

Investor Documentation

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WE RULE. WE GOVERN. WE EMPOWER.

Table of Contents

1. AI ASSISTANT ENHANCEMENT

2. AI AGENTS QUICK START

AI ASSISTANT ENHANCEMENT

AI Assistant Enhancement - SiyaBusa ERP

"The Smartest Thing Ever" - Your First Point of Contact

Updated: January 12, 2026

Status:  Live in Production

Platform: SiyaBusa ERP (Cape Town AWS Region)

What Was Built

1. Comprehensive ERP Knowledge Base (`ERPKnowledgeBase.ts`)

The AI now knows EVERYTHING about the ERP system:

- ****All 25 Integrated Modules**** - Financial, Sales, Purchase, Inventory, HR, Assets, Banking, Treasury, Cash Management, Projects, Manufacturing, Healthcare, Construction, Property, Agriculture, Mining, Logistics, Warehouse, Compliance Hub, Audit Ready, Communications, Proposals, Admin Hub, Practice Management, and Multi-Entity
- ****Module Capabilities**** - What each module can do
- ****Navigation**** - How to get to each feature
- ****Common Questions**** - Pre-built Q&A for each module
- ****Workflows**** - Step-by-step processes (Sales Cycle, Purchase Cycle, Month-end Close, Payroll)
- ****SA Compliance**** - VAT, PAYE, UIF, SDL, POPIA, CIPC
- ****Glossary**** - Business and accounting terms explained simply
- ****Troubleshooting**** - Common problems and solutions

2. Machine Learning System (`AILearningService.ts`)

The AI learns and improves from every conversation:

- **Conversation Storage** - Every interaction is saved for learning
- **Feedback Loop** - Users can mark responses as helpful/unhelpful
- **Pattern Recognition** - Learns successful response patterns
- **FAQ Generation** - Automatically creates FAQs from common questions
- **User Preferences** - Remembers each user's preferred response style
- **Analytics** - Tracks conversation topics, success rates, trends

3. Database Tables (PostgreSQL)

New tables created for AI learning:

```
ai_conversations      -- Stores all conversations for learning
ai_learned_patterns  -- Successful response patterns
ai_faq                -- Frequently asked questions
ai_user_preferences  -- Personalization per user
```

Pre-seeded with **15 common FAQs** about invoices, reconciliation, payroll, VAT, etc.

4. New API Endpoints

Endpoint	Method	Description
/api/ai/feedback/:id	POST	Record feedback (helpful/unhelpful)
/api/ai/faq	GET	Get FAQs for help center
/api/ai/analytics	GET	AI usage analytics
/api/ai/knowledge	GET	Full ERP knowledge base
/api/ai/knowledge/search	GET	Search knowledge base

How The AI Works Now

When User Asks a Question:

1. **Knowledge Search** - First searches the ERP knowledge base for relevant info
2. **Pattern Matching** - Checks learned patterns from past successful responses
3. **Context Loading** - Gets user's recent conversations and preferences
4. **OpenAI Processing** - Sends enriched context to GPT-4o-mini
5. **Response Generation** - Returns helpful, conversational answer
6. **Learning Storage** - Saves conversation for future learning

System Prompt Features:

The AI has been upgraded with:

- Complete module knowledge
- South African compliance awareness
- Step-by-step navigation guidance
- Troubleshooting expertise
- Simple business language (not accounting jargon)
- Proactive suggestions

Example Interactions

User: "How do I create an invoice?" **AI:** "Go to Sales Hub → New Invoice. Select your customer, add line items with quantities and prices, review the totals including VAT, then click Save or Save & Send to email it directly. Would you like me to walk you through it step by step?"

User: "When is VAT due?" **AI:** "VAT returns are due by the 25th of the month following your VAT period. For example, if your period ends on 30 November, your return is due by 25 December. Go to SARS Sentinel to see your upcoming deadlines and prepare your return."

User: "My trial balance doesn't balance" **AI:** "Let's troubleshoot this. Check for: 1) Unposted journal entries, 2) Suspended transactions, 3) Opening balance errors, 4) Rounding issues. Go to Financial Hub → Reports → Trial Balance and look at any accounts with unusual balances. Would you like help identifying the discrepancy?"

Configuration Required

For the AI to work, ensure these environment variables are set:

```
OPENAI_API_KEY=sk-your-key-here
```

The AI uses GPT-4o-mini which costs approximately **R0.50-2 per customer per month** at typical usage.

Files Created/Modified

New Files:

- `backend/src/services/ai/ERPKnowledgeBase.ts` - Comprehensive ERP knowledge
- `backend/src/services/ai/AI LearningService.ts` - Machine learning service
- `backend/migrations/ai-learning-system.sql` - Database tables + seed data

Modified Files:

- `backend/src/services/ai/AIAssistantService.ts` - Enhanced with knowledge + learning
 - `backend/src/routes/ai-assistant.routes.ts` - New learning endpoints
 - `backend/src/controllers/ai-assistant.controller.v2.ts` - New endpoint handlers
-

Analytics Available

Admins can see:

- Total conversations
- Helpful rate (% of positive feedback)
- Top question categories
- Activity trends (last 7 days)
- Learning status

Access via: `/api/ai/analytics`

Next Steps (Optional Enhancements)

1. ****Fine-tuning**** - Train a custom model on successful conversations
 2. ****Voice Interface**** - Add speech-to-text for hands-free use
 3. ****Proactive Alerts**** - AI suggests actions based on system state
 4. ****Multi-language**** - Add Zulu, Afrikaans, Xhosa support
 5. ****Industry Packs**** - Specialized knowledge for healthcare, mining, etc.
-

Support

The AI is designed to be THE FIRST POINT OF CONTACT. Users should:

1. Ask the AI assistant first
2. Check the FAQ section
3. Search the knowledge base
4. Only then escalate to human support

This reduces support load and provides instant, 24/7 assistance.

Live at: <https://siyabusaerp.co.za>

AI AGENTS QUICK START

AI Agents Module - Quick Start Guide

Updated: January 12, 2026

Platform: SiyaBusa ERP (Cape Town AWS Region)

Status:  Integrated

Installation (Complete)

The AI Agents module is already installed and integrated. All you need to do is configure your API keys.

1. Get API Keys

Option A: OpenAI (Recommended)

1. Go to <https://platform.openai.com/api-keys>
2. Create new API key
3. Copy the key (starts with `sk-...`)

Option B: Anthropic (Alternative)

1. Go to <https://console.anthropic.com/>
2. Create API key
3. Copy the key (starts with `sk-ant-...`)

Option C: Both (Fallback support)

You can configure both. OpenAI will be used first, Anthropic as fallback.

2. Configure Environment

Add to your `.env` file:

```
# Required: At least one API key
OPENAI_API_KEY=sk-your-openai-key-here
# OR
ANTHROPIC_API_KEY=sk-ant-your-anthropic-key-here

# Optional: Configuration
AI_DEFAULT_MODEL=gpt-4
AI_MAX_TOKENS=2000
AI_TEMPERATURE=0.7
```

3. Deploy Database Schema

```
# For Cape Town ECS deployment, connect to RDS directly:
# Host: worldclass-erp-db.c92ou2c2e43l.af-south-1.rds.amazonaws.com
# Database: erp_database

# Navigate to backend
cd /workspaces/WorldClass-ERP/backend

# Run migration
psql $DATABASE_URL -f database/migrations/018_ai_agents_module.sql

# Verify
psql $DATABASE_URL -c "SELECT agent_name FROM ai_agents;"
```

Expected output:

```
      agent_name
-----
Sales Assistant
Procurement Assistant
Finance Assistant
HR Assistant
Compliance Assistant
Analytics Assistant
Treasury Assistant
Inventory Assistant
Logistics Assistant
(9 rows)
```

4. Restart Backend

```
# For ECS deployment, redeploy the service:
aws ecs update-service --cluster worldclass-erp-cluster --service worldclass-erp-backend
```

Look for:

```
 Server running on port 3000  
 ERP System initialized
```

5. Test AI Agents

Test 1: List Agents

```
curl -X GET https://siyabusaerp.co.za/api/ai/agents \  
-H "Authorization: Bearer YOUR_JWT_TOKEN"
```

Should return 9 agents.

Test 2: Chat with Sales Assistant

```
curl -X POST https://siyabusaerp.co.za/api/ai/agents/SALES_ASSISTANT/chat \  
-H "Authorization: Bearer YOUR_JWT_TOKEN" \  
-H "Content-Type: application/json" \  
-d '{  
  "message": "What are the top 3 tips for closing deals?"  
'
```

Should return AI response with conversation ID.

Test 3: Get Suggestions

```
curl -X GET https://siyabusaerp.co.za/api/ai/suggestions \  
-H "Authorization: Bearer YOUR_JWT_TOKEN"
```

6. Frontend Integration

Add AI chat component to your frontend:

```
// Example: Chat with any agent
async function chatWithAgent(agentCode: string, message: string) {
  const response = await fetch(`/api/ai/agents/${agentCode}/chat`, {
    method: 'POST',
    headers: {
      'Authorization': `Bearer ${token}`,
      'Content-Type': 'application/json'
    },
    body: JSON.stringify({ message })
  });

  return await response.json();
}

// Usage
const result = await chatWithAgent('SALES_ASSISTANT', 'Help me create a quotation');
console.log(result.response);
```

Available Agents

Agent Code	Module	Purpose
SALES_ASSISTANT	Sales	Quotations, forecasting, customer insights
PROCUREMENT_ASSISTANT	Purchase	PO optimization, supplier analysis
FINANCE_ASSISTANT	Financial	Reconciliation, variance analysis
HR_ASSISTANT	HR	Onboarding, payroll, policies
COMPLIANCE_ASSISTANT	Compliance	Risk assessment, deadlines
ANALYTICS_ASSISTANT	Reports	Data insights, trends
TREASURY_ASSISTANT	Treasury	Cash flow, FX, investments
INVENTORY_ASSISTANT	Inventory	Stock optimization, demand forecasting
LOGISTICS_ASSISTANT	Logistics	Route optimization, fleet tracking

Common Use Cases

Sales: Generate Quotation

```
POST /api/ai/sales/generate-quotation
{
  "customerId": "cust-123",
  "items": [{"productId": "prod-456", "quantity": 10}]
}
```

Compliance: Check Status

```
GET /api/ai/compliance/check
```

Finance: Explain Variance

```
GET /api/ai/finance/explain-variance?accountCode=4000&period=2024-01
```

Sales: Forecast

```
GET /api/ai/sales/forecast?period=quarter
```

Monitoring

Check Usage

```
SELECT
  a.agent_name,
  SUM(ua.total_messages) as messages,
  SUM(ua.total_tokens) as tokens
FROM ai_usage_analytics ua
JOIN ai_agents a ON ua.agent_id = a.agent_id
WHERE ua.usage_date >= CURRENT_DATE - INTERVAL '7 days'
GROUP BY a.agent_name;
```

View Recent Conversations

```
SELECT
  a.agent_name,
  c.conversation_title,
  c.last_message_at,
  (SELECT COUNT(*) FROM ai_messages WHERE conversation_id = c.conversation_id) as message_count
FROM ai_conversations c
JOIN ai_agents a ON c.agent_id = a.agent_id
ORDER BY c.last_message_at DESC
LIMIT 10;
```

Troubleshooting

Error: "No AI provider configured"

Solution: Add `OPENAI_API_KEY` or `ANTHROPIC_API_KEY` to `.env` and restart backend

Error: "Agent not found"

Solution:

```
# Check if agents exist
psql $DATABASE_URL -c "SELECT * FROM ai_agents WHERE is_active = true;"

# If empty, re-run migration
psql $DATABASE_URL -f database/migrations/018_ai_agents_module.sql
```

Slow responses

Solution:

- Switch to `gpt-3.5-turbo` (faster, cheaper)
- Use streaming endpoint: `/api/ai/agents/:agentCode/stream`

High costs

Solution:

- Reduce `AI\_MAX\_TOKENS` in `.env`
- Set monthly limits per tenant
- Use gpt-3.5-turbo instead of gpt-4

Next Steps

1.  ****Module Complete****: AI Agents fully integrated
2.  ****Multi-Entity****: Add multi-entity support (next task)
3.  ****Deployment Packages****: Create deployment packages for all new modules
4.  ****Testing****: Comprehensive testing with user

Cost Estimation

OpenAI GPT-4 Pricing (as of Nov 2024):

- Input: \$0.03 per 1K tokens
- Output: \$0.06 per 1K tokens

Example: 100 conversations/day with 500 tokens each = \$3/day = \$90/month

Optimization:

- Use GPT-3.5-Turbo: ~10x cheaper
- Cache common queries
- Limit token usage per query

Support

Full documentation: `/backend/AI-AGENTS-DOCUMENTATION.md`

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Live URL: <https://siyabusaerp.co.za>